

The Guarded Agent

Autonomous options paper trading where AI proposes and deterministic controls decide.

PAPER ONLY / EXECUTION OFF
KILL SWITCH ON

\$100K
VERIFIED EQUITY

LEVEL 3
OPTIONS APPROVAL

0
ORDERS AND FILLS

2
PRODUCTION STRUCTURES

128
TESTS PASSING

01 / PROBABILISTIC LAYER

AI reasons - it never brokers

- Daily direction plus a one-minute pullback/reversal gate limits AI calls.
- Featherless receives only sanitized, timestamped Alpaca evidence.
- One strict JSON decision: NO_TRADE or one allowlisted options proposal.
- Malformed, stale, missing, or unsupported output fails closed to abstention.
- Evidence, model metadata, hashes, token use, cost, and rejected alternatives are audited.
- The AI module has no Alpaca order client and cannot change risk limits.

02 / DETERMINISTIC LAYER

Capital protection is recomputed

- Production permits only signal-matched call or put debit spreads.
- Debit and maximum loss are rebuilt from option legs, never trusted from AI output.
- Gates cover confidence, 2--10 DTE, quote freshness, spread, OI, trade risk, portfolio risk, drawdown, and positions.
- Existing positions block new entries for exit review; profit, stop, 15-minute hold, expiry, partial-fill, and assignment states use AI-independent policies.
- Two execution approvals plus an independent kill switch guard the paper broker boundary.

03 / ALPACA INFRASTRUCTURE

Real APIs, read-only proof first

- Official alpaca-py clients reconcile account, clock, orders, positions, contracts, and market data.
- Alpaca MCP evidence covers option chain, quote, IV, and Greeks on the free indicative feed.
- Two immutable 22-contract SPY snapshots include timestamps, Greeks, schema checks, and SHA-256.
- Closed-market live shadow evidence proves NO_TRADE and order_sent=false without an AI call.
- Worker and dashboard images bake in fail-closed paper defaults; CI is configured to build them.

04 / DIFFERENTIATOR AND EVIDENCE

Auditable abstention beats bravado

- A hash-chained trace links evidence, AI thesis, recomputed risk, preview, and broker state.
- Real-source shadow P&L uses adverse quote sides and exact legs; missing marks stay unmarked.
- Seeded scenarios and negative underlying holdouts are disclosed without being called options alpha.
- Current limitation: multi-date option history and open-market AI shadow evidence are still pending.
- Target users are small systematic traders and fintech builders; a future path is a hosted risk-and-audit API.
- Every failure mode becomes NO_TRADE, risk rejection, manual review, or blocked execution.

EXIT REVIEW -> ALPACA EVIDENCE -> AI PROPOSAL -> RISK GATES -> UNSENT PREVIEW

No provider failure, stale quote, rejected proposal, or audit error can become an order.

CURRENT FIRST-ORDER DECISION: NO-GO

Execution stays disabled until multi-session validation, image smoke tests, durable cloud audit, and explicit user approval. Live trading is prohibited by configuration and project policy.